

Compiled: converts source code directly into machine code.

Interpreted: executes source code line-by-line using an interpreter without pre-compiling.

C#: it is a hybrid language. First, the C# compiler compiles the code into Intermediate Language (IL), then the JIT compiler compiles this intermediate code into machine code at runtime.

---

Casting types:

1)Implicit: an automatic conversion from lower range datatypes into higher range ones.

2)Explicit: a manual conversion when implicit conversion is not possible (e.g., converting from double to int, which may cause data loss).

3)Convert: converts between various base types and safely handles null values (returns 0 for null).

4)Parse: converts a string representation of a number into its target datatype, but throws an exception if the string is invalid or null.